Manoj Babu **Katragadda**

Principal Engineer

Remote · Asia

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Principal Engineer building reusable data and ML platforms. Hands-on expertise in distributed systems, database internals, Spark connectors, and performance optimization. Seeking full-time roles with long-term ownership.

Experience

Zerotoone AI

Remote · Asia

Principal Platform Engineer

Apr 2025 – Apr 2026

- Built **Zest, an MLOps platform** with **multi-tenant workspace isolation** and a declarative Python SDK covering the ML lifecycle from data segmentation to model deployment. Its AST-based compiler generates Spark jobs, resolves resource dependencies, and versions artifacts. **[Case Study]**
- Designed a **composable PySpark ETL framework** for feature, label, and training-data pipelines, reducing total code per comparable pipeline by **75%**. Reusable hooks, pluggable extractors and writers, and Write-Audit-Publish quality gates standardize execution across environments. **[Case Study]**
- Reduced the runtime of a daily Spark MERGE job from **60 minutes to 4 minutes** on a **2-billion-row Apache Iceberg table** using storage partition joins and shuffle hash join hints. **[Case Study]**
- Deployed Spark on Kubernetes with S3-backed shuffle and spot executors, reducing **Spark compute costs by 90%**. Individual production ETL jobs processed **10 TB+ of input per run**. **[Case Study]**
- Extended **Airflow** for Kubernetes job orchestration and DAG delivery: Ray operators with fault-tolerant reattachment, asynchronous Redis triggers, and hot-reloaded DAG bundles that avoid Helm upgrades and pod restarts. **[Case Study]**

AirAsia Berhad · 4 yrs 5 mos

Kuala Lumpur, Malaysia

Principal Engineer (2 yrs)

Sep 2019 – Feb 2024

- Refactored **Talaria's ingestion and flush engines** around shared block/row encoding, separating internal storage from connector-level conversion for formats such as **Parquet and ORC**.
- Developed and open-sourced Talaria's **Apache Spark connector** in Java using **DataSource V2**: gRPC reads, vectorized columnar batches, and checkpointed micro-batch streaming; enabled processing of Talaria data in Spark and writing the results to Apache Iceberg.
- Built Talaria integrations for **Google Cloud Pub/Sub, BigQuery, and Google Cloud Storage**: real-time event ingestion with exactly-once delivery through Pub/Sub, and data export to BigQuery and GCS.
- Implemented a **data mesh** using Airflow for orchestration, Spark and Apache Iceberg for lakehouse processing and storage, Trino and Cube.js for self-service analytics, and Backstage for governance.
- Built the **experimentation platform** with a Go CQRS framework that compiles experiment definitions into Lua, an event-sourcing foundation for complex event processing, and a streaming framework that feeds evaluation data into the statistics engine. Worked with product managers to define experiments and evaluation criteria.
- Created reusable deployment code for **more than 10 Airflow systems on Kubernetes**, each running **over 200 jobs daily**, replacing per-team single-node setups. Built Terraform modules, Kubernetes manifests, and GitLab CI templates for Google Kubernetes Engine.
- Mentored a staff data engineer building a Snowplow-inspired **Python ETL framework** using medallion architecture and a senior software engineer building a **Backstage governance portal** for access policies, JSON Schema definitions for events, A/B experiments, and JWT issuance.
- Built a NestJS taxi queue service using onion architecture and established its framework as the foundation for legacy Node.js service migrations. Decoupled business entities from persistence to support migration between MongoDB and Firestore; implemented Redis locking for caching and event-driven messaging via a **Pub/Sub controller**.

Experience (continued)

Honestbee
Singapore; Bangalore, India

Data Engineer II
Jun 2018 – Mar 2019

- Built ETL pipelines using Airflow, Spark, AWS Glue, and **Redshift**, including a custom Airflow operator for AWS Glue jobs.
- Worked with data scientists to schedule model runs in Airflow and deploy model-serving APIs on Kubernetes.

Zivame · 2 yrs 9 mos
Bangalore, India

Senior Software Engineer (2 yrs)
Aug 2015 – May 2018

- Developed **Spark-based product recommendations** using customer preferences and behavior.
- Led the product-search migration from Solr to **Elasticsearch**; implemented Nginx caching to improve website performance.
- Built product discovery features and supporting **Java APIs** for filtering, sorting, search, and autocomplete; centralized service logs and analytics with the **ELK stack**.

Kore AI
Hyderabad, India

Software Engineer
Sep 2014 – Jul 2015

- Developed Node.js backend APIs.

Technical Skills

Data platforms	:	Spark · Apache Iceberg · Trino · BigQuery · Redshift · Postgres · Talaria · lakeFS · Cube.js · Data Lakes · Data Mesh
Orchestration	:	Airflow · Prefect
Cloud & infrastructure	:	GCP · AWS · Kubernetes · Docker · Terraform
Languages & libraries	:	Go · Python · Java · Rust · TypeScript · Node.js · React · Lua
Engineering	:	Data · DevOps · Infrastructure · Backend · Experimentation · Open Source · SOLID
Developer tooling	:	*Nix · Zsh · tmux · Neovim

Open Source & Speaking

- Contributor to and maintainer of **Talaria**, an open-source time-series database in Go. Contributions span database internals, Parquet/ORC handling, and Google Cloud integrations.
- Authored and open-sourced the Java **Apache Spark connector for Talaria**, enabling batch and stream processing of Talaria data in Spark.
- Repaired cache and prefetch deadlocks in a fork of the **Spark S3 shuffle plugin**; added bounded prefetch waits, Spark 4.0/Scala 2.13 support, and reliable-storage declarations for executor decommissioning.
- Built **Bob**, an open-source voice interface for AI coding agents. Integrated Pi, OMP, and Claude with local speech processing, MCP profiles, and durable background jobs using TypeScript, NestJS, and React.
- Presented a talk on implementing prefect-spark-on-k8s-operator at **PyCon Malaysia 2023**.
- Authored open-source Prefect integrations for **Spark-on-Kubernetes job scheduling** and **lakeFS housekeeping**, using the Spark operator and lakeFS API.

Education

Indian Institute of Technology Kharagpur
IIT Kharagpur, India

Bachelor of Technology
Jul 2009 – May 2014